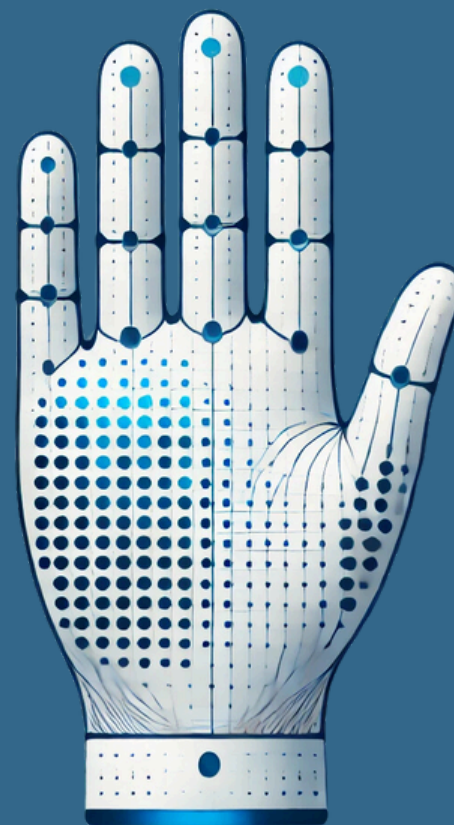




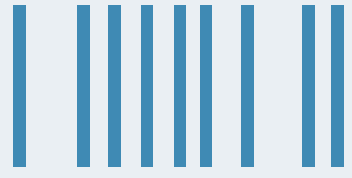
HDSENSE GLOVE

Redefining Tactile feedback with Compressed Sensing

By Harish Balasubramanian

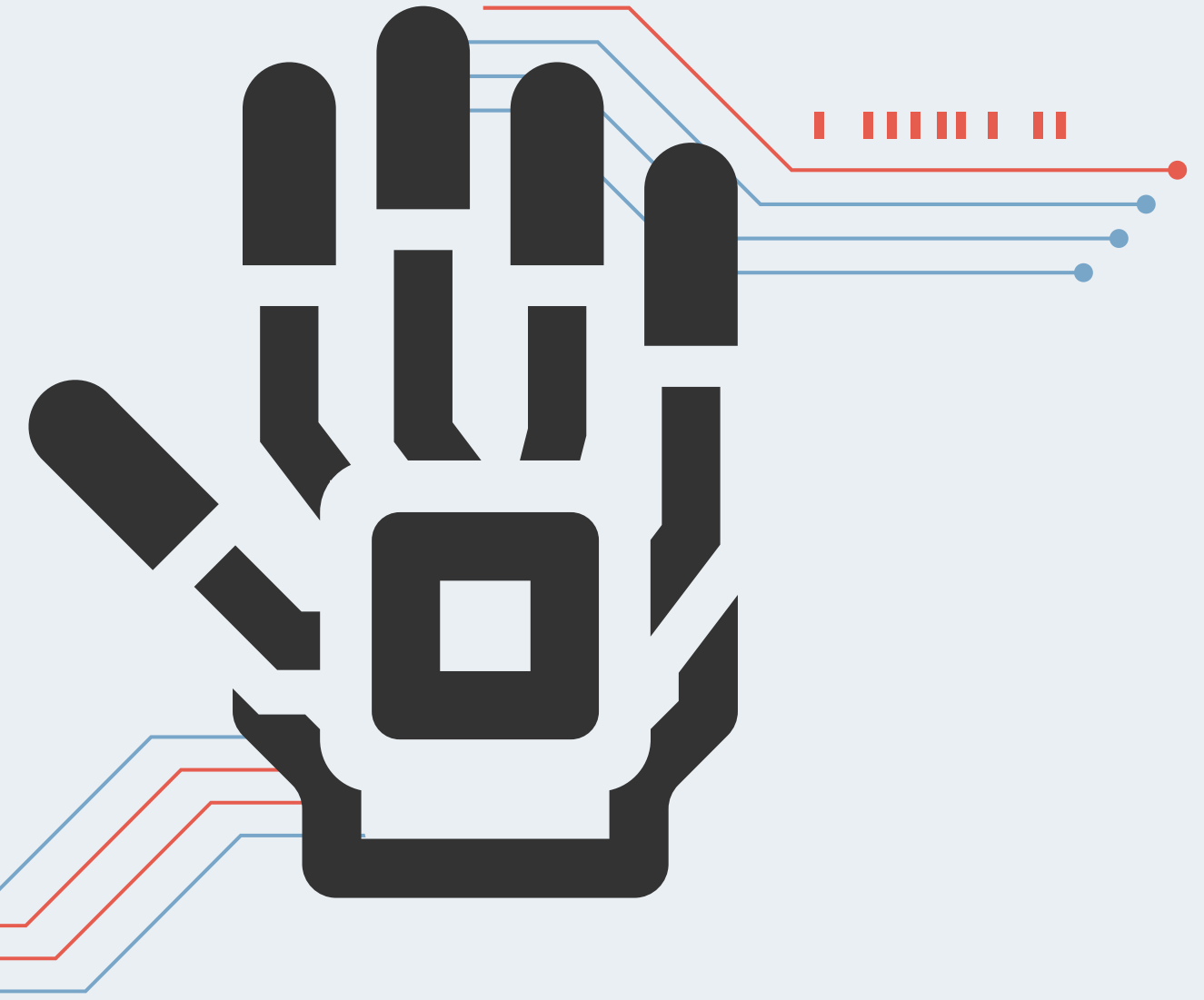
Graduate Student, Biomedical Engineering, Johns Hopkins University

*Under the guidance of Dr. Nitish Thakor and Arik Slepian
(Neuroengineering and Biomedical Instrumentation Lab)*



THE PROBLEM

- Current Gloves have limited sensor density and lack real-time capabilities.
- Need for high-speed scalable and precise in prosthetics, robotics, sports, and more.
- Data overload challenges with high density systems.



THE INNOVATION

»» 1024 Sensors

32x32 High Density
Sensor Grid

»» Compressed Sensing

Venus is the second
planet from the Sun

»» Real-time

20ms detecting
speed after contact

»» 88% accuracy

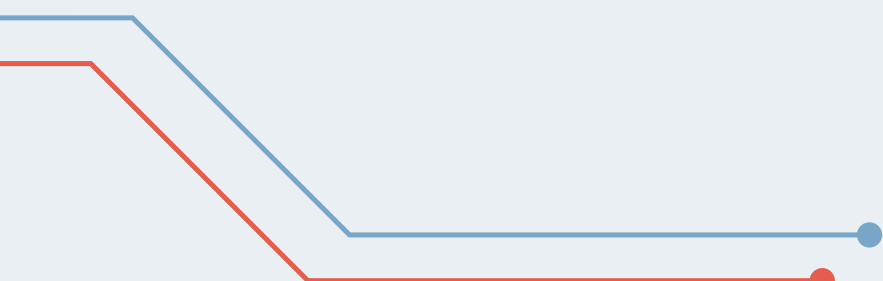
Detecting objects

»» Low power

Low powered Teensy
Board

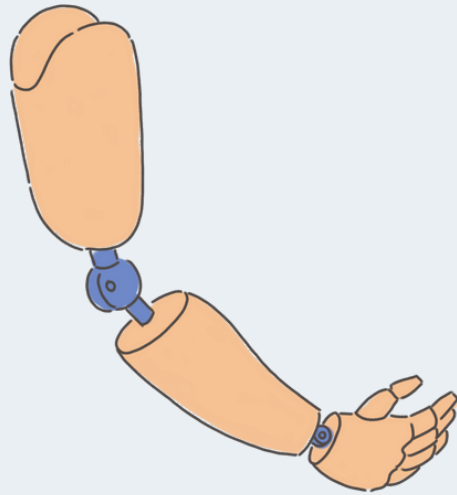
»» 18X faster

Subsampling method*
over Raster scanning



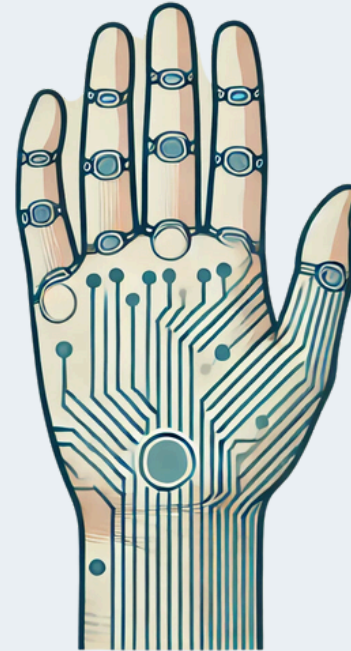
**Slepyan et al., 2024.*

MARKET ANALYSIS



Upper Limb Prosthetics

1.1 Billion* in 2021,
growing at 5.4% CAGR



Electronic Skin

19.6 Billion* by 2029



Smart Sports Equipment

2.6 Billion* in 2021

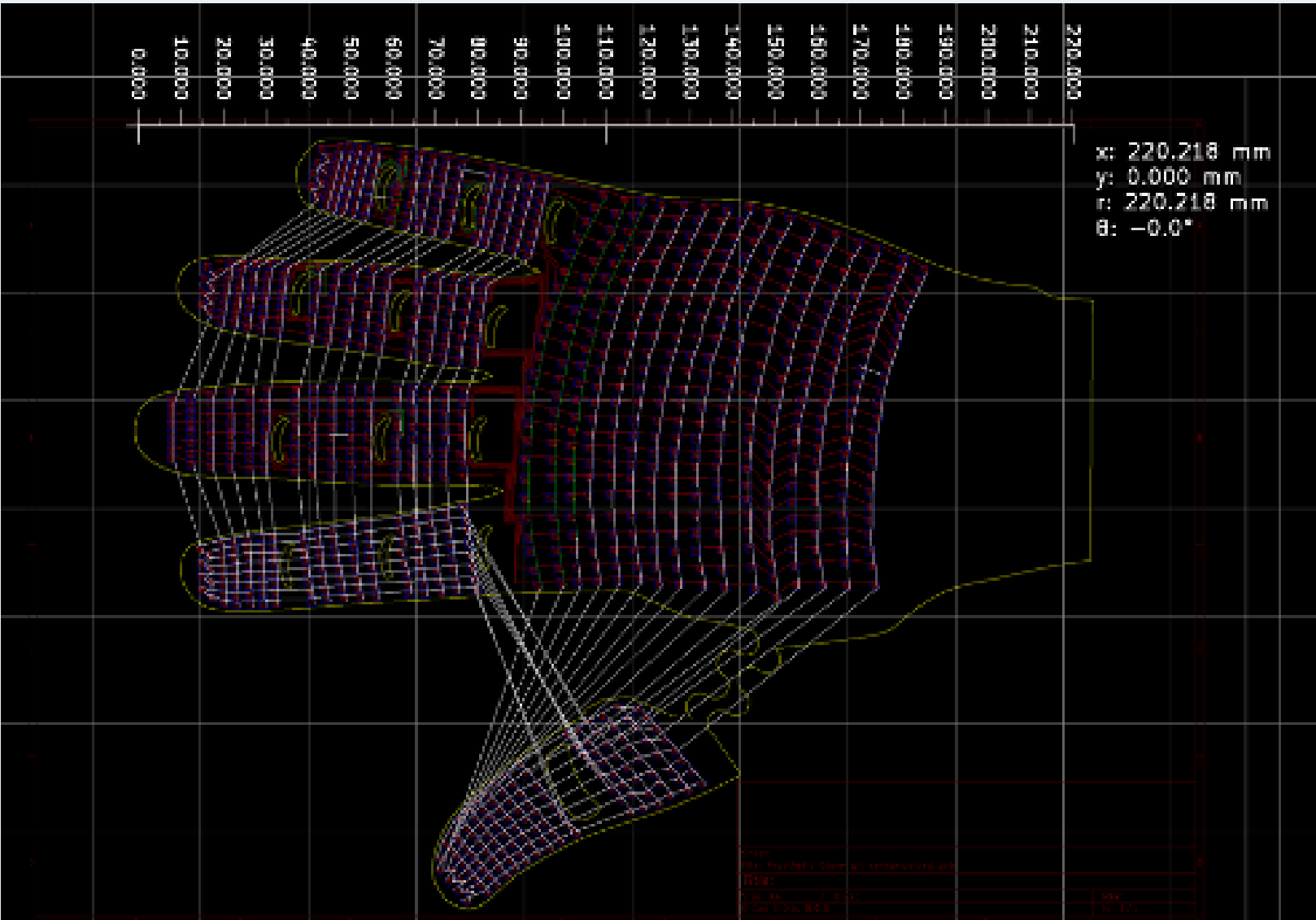
**Data From GrandView Research*

COMPETITIVE EDGE

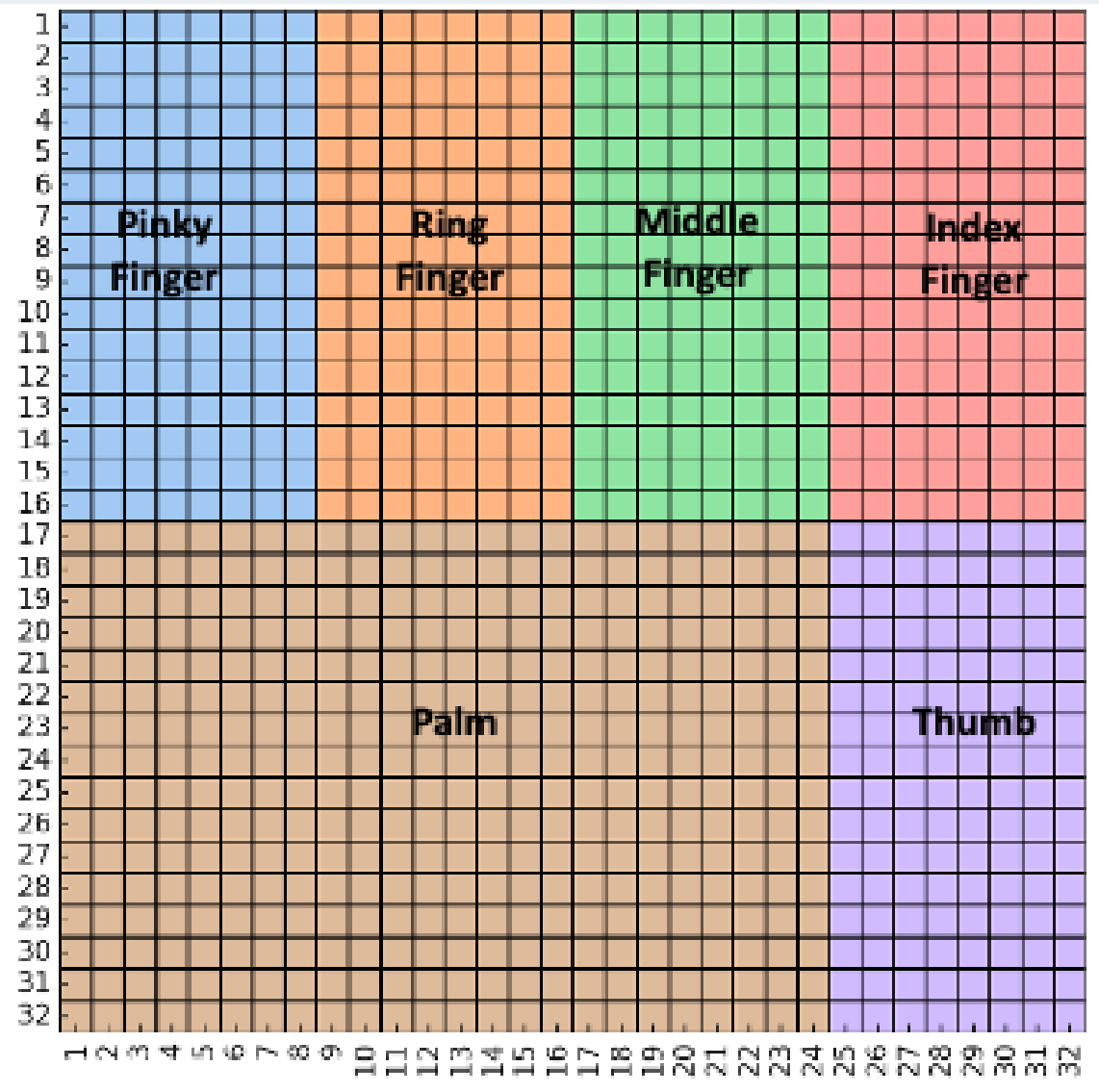
FEATURE/ METRIC	STAG GLOVE(MIT)	ELECTRONIC SKIN SYSTEMS	SMART SPORTS GLOVE	HDSENSE GLOVE
Sensor Density	548	Low Density	Low Density	1024
Frame Rate	7.3Hz	<100Hz	<100Hz	1000Hz
Latency	High	High	Medium	20ms
Energy Consumption	Medium	Low	Low	Low
Resolution	Medium	Low	Low	Very High



DESIGN AND DEVELOPMENT



Design for Average Male Left Hand



32x32 Pixel map for tactile glove

PATENT STATUS

Patent for Compressed Sensing Technology using **ACTS** and Machine Learning is **pending approval** and **refinement**.

Key Claims:

- **High-Speed Tactile Sensing:**
Integration of compressed sensing to reduce data overload and latency.
- **Machine Learning-Driven Reconstruction:**
Uses a learned tactile dictionary for precise and efficient signal reconstruction.
- **Scalable and Cost-Effective:**
Adaptable for multi-industry applications in prosthetics, robotics, and sports.
-

PRICING

Cost



Production Cost:
80 US Dollars

Selling Cost:
160 US Dollars

Competition



250-500 US Dollars
For slower speeds and bulkier design

Cost vs Benifit



Exceeds in tactile resolution and
speed for more affordable price



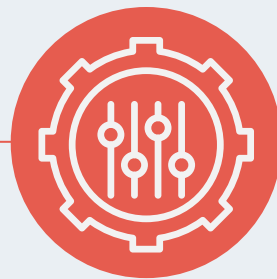
WHERE DO WE GO FROM HERE?



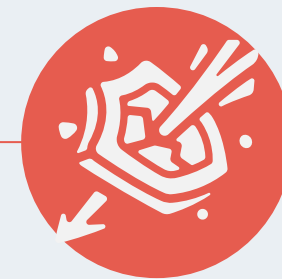
Testing and
Refining Prototype



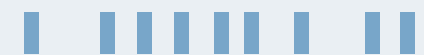
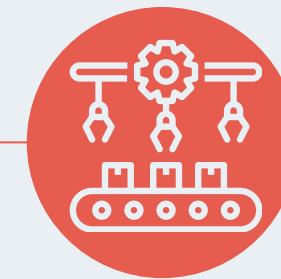
Customer Feedback



Custom Glove For
the sensor PCB



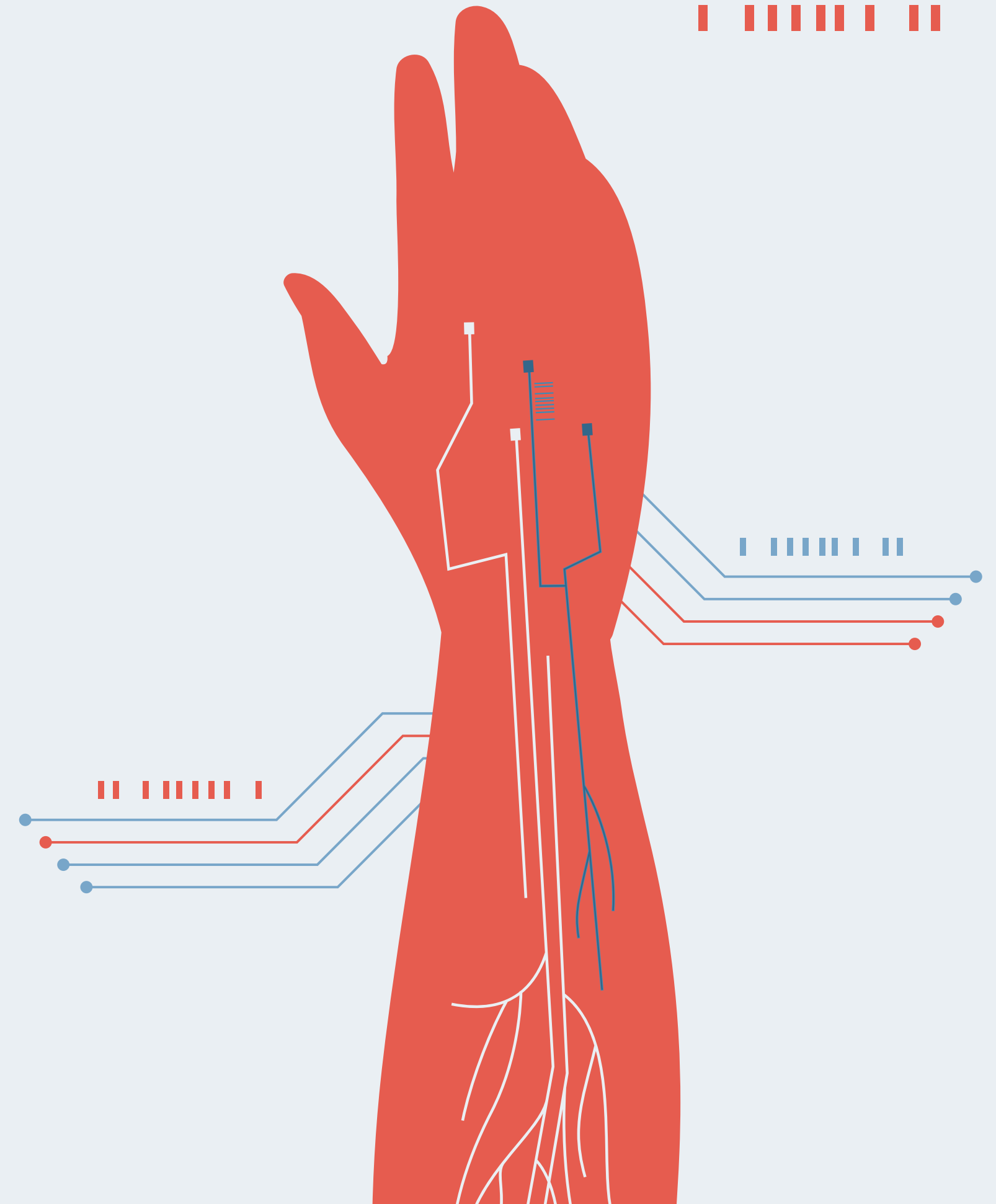
Manufacturing
and Scale



CONCLUSIONS

Why HDSense Glove?

Address Critical Gaps in Tactile Sensing
Multi-Industry Applications
Scalability and Potential Market for Growth



THANKS

Do you have any questions?



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